

RAKSHITH RAJ M

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PROFESSIONAL SUMMARY

MLOps and AI Engineer with hands-on experience building end-to-end ML and LLM systems – from model training and experiment tracking (MLflow, DVC) to production deployment (Docker, FastAPI, CI/CD) and observability (Prometheus, Grafana, Langfuse). Skilled in Retrieval-Augmented Generation, vector search, and local LLM inference, backed by a full-stack foundation in React, Node.js, TypeScript, and AWS.

PROFESSIONAL EXPERIENCE

Software Developer (SDE) **August 2025 – Present**
GPUNET, India

- Architecting and developing scalable SaaS applications using ReactJS and TypeScript
- Implementing cloud-native solutions on AWS with emphasis on performance optimization and reliability
- Designing optimized MySQL database schemas and RESTful APIs for high-performance data operations
- Establishing Git workflows, branching strategies, and CI/CD pipelines for efficient team collaboration
- Integrating third-party services and APIs to enhance application functionality and user experience

DevOps Engineer Intern **February 2025**
IBM / Rooman Technologies, Bengaluru *On-the-Job Training*

- Completed intensive AI-DevOps Engineer training program certified by Visvesvaraya Technological University
- Gained hands-on experience with containerization (Docker), orchestration (Kubernetes), and cloud platforms
- Implemented CI/CD pipelines for automated testing, building, and deployment of applications
- Worked with infrastructure as code, monitoring tools, and automation frameworks for production environments
- Applied DevOps best practices including version control, configuration management, and continuous integration

KEY PROJECTS

- SupportForge – RAG-Powered Telegram Support Bot** | *Python, FastAPI, LangGraph, Ollama, Qdrant, PostgreSQL*
- Built a production-style RAG support bot with LangGraph orchestration over locally-hosted LLMs (Ollama), removing external API dependencies
 - Implemented Qdrant vector search (768-dim nomic-embed-text embeddings) with top-5 similarity retrieval over ingested PDF, Markdown, and text documents
 - Designed an automated response-quality pipeline using DeepEval (faithfulness, hallucination) and Ragas (context precision/recall) to continuously monitor RAG output
 - Built versioned prompt management for safe prompt promotion without redeployment, plus persona-based incident simulation to surface knowledge gaps
 - Instrumented full observability with Prometheus/Grafana (latency, token usage) and Langfuse (LLM tracing); used Celery/Redis for async background processing

- Credit Card Fraud Detection – End-to-End MLOps Pipeline** | *Python, Scikit-learn, MLflow, DVC, FastAPI*
- Built an end-to-end MLOps pipeline for fraud detection on a severely imbalanced dataset (0.173% fraud rate), optimizing for precision/recall/PR-AUC over accuracy
 - Trained a balanced-weight Random Forest classifier with modular training, inference, and pytest-based testing components
 - Tracked experiments and versioned datasets/model artifacts using MLflow and DVC for full reproducibility
 - Served predictions via a Dockerized FastAPI inference API with health and status endpoints
 - Automated a GitHub Actions CI pipeline to train and validate models on every commit without external dependencies

- container-lab – Optimized FastAPI Deployment Pipeline** | *Python, FastAPI, SQLAlchemy, Docker, Alpine Linux*
- Designed a two-stage Docker build that compiles native wheels (uvloop, httptools) in a builder stage and ships only the runtime venv, cutting the production image to roughly 121MB
 - Hardened the container for production with non-root user execution and a built-in `/health` HEALTHCHECK endpoint
 - Built a FastAPI + SQLAlchemy backend with CRUD APIs and Swagger docs, backed by volume-persisted SQLite for stateful deployments

TECHNICAL SKILLS

AI / Machine Learning: TensorFlow, Scikit-learn, Deep Learning, LLM Integration, RAG
MLOps & Observability: MLflow, DVC, Qdrant, LangGraph, Celery, Redis, Prometheus, Grafana, Langfuse, DeepEval, Ragas
Cloud & DevOps: AWS, Docker, Kubernetes, CI/CD Pipelines, Loki, Linux
Programming Languages: JavaScript, TypeScript, Python, SQL
Backend Development: Express.js, Hono, Bun, RESTful API Development, Microservices Architecture
Frontend Development: HTML, CSS, React.js, Responsive Web Design
Databases: MySQL, NoSQL, PostgreSQL, SQLite
Developer Tools: Git, Version Control, Jupyter Notebook, Selenium
Data & Analytics: Data Analysis, Data Visualization